

Video Tutor Agent

Paste a YouTube link, then **talk** to a tutor that answers only from the video, in English, Hindi, Hinglish or other Indian languages. Built entirely on **Sarvam** (LLM, speech-to-text, text-to-speech).

Scope

- **In scope:** YouTube link → transcript; a **hands-free voice conversation** you can interrupt anytime; answers grounded in the video with **clickable timestamps**; Indian-language and code-mixed speech; web UI.
- **Out of scope (for now):** videos over 60 min / RAG; understanding what is shown on screen (slides, diagrams); accounts, database, deployment.

High-level architecture

- **Transcript builder:** **yt-dlp** downloads audio + captions → **Sarvam saaras:v4** batch speech-to-text (codemix, key terms from the channel and tags) → sentence-level timestamps, sharpened with caption timing; gaps filled from captions → cached. **~40 s** for a 9-min video.
- **Voice session** (WebSockets): mic → **Sarvam realtime STT** (detects end of speech) → **turn analyzer** (small, fast LLM call: question / unfinished / "stop" / not for the tutor) → **sarvam-105b** with the full transcript, streamed → **Sarvam streaming TTS**, sentence by sentence. **~1.2–1.5 s** from the student going quiet to hearing the tutor.
- **Web UI** (plain HTML/JS): YouTube player, searchable transcript, live captions, timestamp chips that jump the video, hold-to-talk fallback.

Challenges & solutions

- **Slow answers:** the LLM's reasoning mode took **14 s** and sometimes returned nothing → turned off: **0.5–0.9 s**.
- **Cut off while thinking:** fixed silence timers ended turns mid-sentence → the turn analyzer judges whether the thought is **finished by meaning** and waits for unfinished ones.
- **Background chatter and music:** family talk got answered; a song froze the tutor for 18 s → speech not meant for the tutor is **ignored**; the tutor **lowers its volume** instead of stopping until it is sure; "hmm / haan" never interrupt; "ruko / wait" stops it.
- **Outside facts and wrong timestamps:** strict "video only" rules, a citation on every factual sentence, and a **citation check** that snaps each timestamp to a real transcript line.
- **Poor Hinglish transcripts:** YouTube captions wrote English terms in Hindi script; **saaras:v3 silently dropped 2 minutes** → saaras:v4 in codemix mode plus a completeness check against captions.
- **Language switching:** a Hindi question got an English answer → the reply language is set **per question**.

Results

Turn analyzer **20/21** correct on labelled speech (~200 ms). End-to-end voice scenarios **8/8** (thinking pause, chatter, "hmm", "ruko", barge-in, grounding).